

NAME: _____

PERIOD: _____

Fourteen Caption Users

AP Statistics · Unit 3 · Topic 3.2

[STAR] THE PROBLEM

Suppose a streaming platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\hat{p} = 0.14$.

Jules: “Because $n = 100 > 30$, $\hat{p}$ is approximately normal. I get $P(\hat{p} \geq 0.14) = 0.0135$, which proves caption use increased.”

Nora: “Because $np = 8 < 10$, a normal model is not justified. Therefore $\hat{p}$ is biased and this sample tells us nothing.”

A binomial calculator gives $P(X \geq 14) = 0.0282$ for $X \sim \text{Bin}(100, 0.08)$. This closely approximates sampling from 5,000 accounts.

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE — one sentence before you work the parts

Does finding 14 caption users instead of the expected 8 prove that use increased? Give one reason.

[BOOK] CHECK INDEPENDENCE, THEN SHAPE (keep on the page)

For an SRS, $\mu_{\hat{p}} = p$ and $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$ when $n < 10\%$ of the population makes without-replacement dependence negligible. Approximate normality is a separate question: require both $np \geq 10$ and $n(1-p) \geq 10$; a generic $n > 30$ rule is not enough for proportions. If Large Counts fails, use an exact binomial calculation when independence is reasonable or simulate the sampling process.

Work (a)–(c).

DOK 1 (a) Identify p and verify the sample proportion $\hat{p} = 14/100$.

DOK 2 (b) Find the mean and standard deviation of the sampling distribution of $\hat{p}$. Check the 10 percent condition and both Large Counts values.

DOK 3 (c) In 3–4 sentences, explain what each student gets right and wrong. Use the condition checks and the supplied probability 0.0282. State what the sample suggests about caption use and why a failed normal-shape check does not make the estimate biased.

Frame: The condition _____ passes, but _____ fails.

Frame: The probability 0.0282 suggests _____; failure of normal shape does not mean _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.